四川大学期末考试试题（闭卷）

（2021~2022 学年第 2 学期） A 卷

课程号-课序号：311232030 / 01~06 课程名称：软件工程导论 任课教师：______

适用专业年级：软件工程 2020 级 学生人数：286 印题份数：300 学号：______ 姓名：______

考生承诺

我已认真阅读并知晓《四川大学考场规则》和《四川大学本科学生考试违纪作弊处分规定（修订）》，郑重承诺：

1. 已按要求将考试禁止携带的文具用品或与考试有关的物品放置在指定地点；
2. 不带手机进入考场；
3. 考试期间遵守以上两项规定，若有违规行为，同意按照有关条款接受处理。

考生签名：

题 号	1 (10%)	2 (12%)	3 (10%)	4 (16%)	5 (12%)	6 (10%)	7 (20%)	8 (10%)
得 分								
卷面总分			阅卷时间					

注意事项：1. 请务必将本人所在学院、姓名、学号、任课教师姓名等信息准确填写在试题纸和答卷纸上；
2. 请将答案全部填写在本试题纸上；
3. 考试结束，请将试题纸、答卷纸、添卷纸和草稿纸一并交给监考老师。

评阅教师	得分

1. (10 Points) Although prototyping can be used as a process model, it is more commonly used as a technique in software development. Please answer following questions:

(1) How can you use figure to illustrate the prototyping model?

(2) What is the role of the prototyping in requirement engineering?

评阅教师	得分

2. (12 Points) What is functional independence? What are the two qualitative criteria to assess independence? Please give the definitions of these two criteria and descript the goal of functional independence in software design.

评阅教师	得分

3. (10 Points) Please draw a Use Case Diagram in UML for an application of "Retail System (零售系统)" according to the following description.

We will develop a retail system for company A, through which buyers and sellers will conduct transactions. Customers send and track orders, while company A is responsible for shipping goods and notifying customers to pay.

评阅教师	得分

4. (16 Points) Please draw an Activity Diagram in UML for the use case "ACS-DCV", according to the following description.

Use case: Access camera surveillance via the Internet—display camera views (ACS-DCV)

Actor: homeowner

If I'm at a remote location, I can use any PC with appropriate browser software to log on to the SafeHome Products website. I enter my user ID and two levels of passwords and once I'm validated, I have access to all functionality for my installed SafeHome system. To access a specific camera view, I select "surveillance" from the major function buttons displayed. I then select "pick a camera" and the floor plan of the house is displayed. I then select the camera that I'm interested in. Alternatively, I can look at thumbnail snapshots from all cameras simultaneously by selecting "all cameras" as my viewing choice. Once I choose a camera, I select "view" and a one-frame-per-second view appears in a viewing window that is identified by the camera ID. If I want to switch cameras, I select "pick a camera" and the original viewing window disappears and the floor plan of the house is displayed again. I then select the camera that I'm interested in. A new viewing window appears.

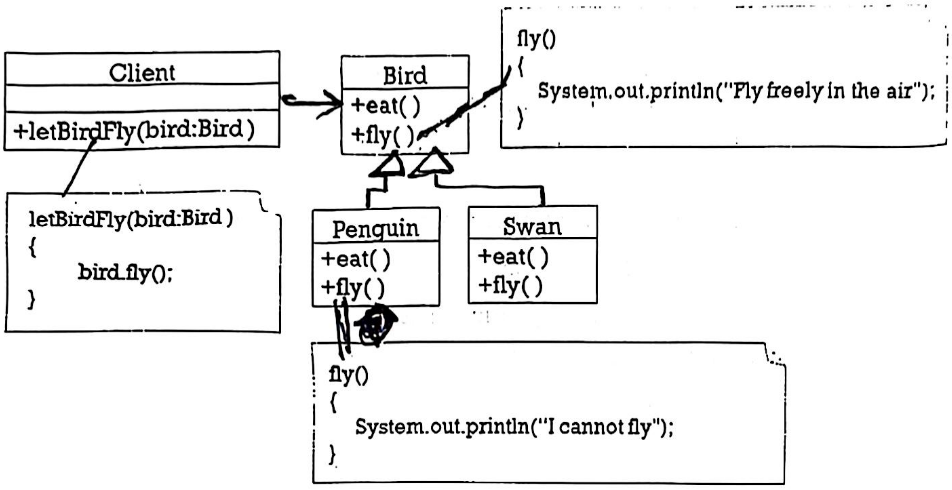
评阅教师	得分

5. (12 Points) Please design a GUI for a Logistics System (物流系统) according to the Golden Rules. Following is the description of system.

Kitty is an employee of the logistics company, mainly responsible for receiving and sending parcels. The receiving business needs this information: weight, sender's name and contact information, recipient's name and contact information, and addresses of both parties. Kitty gets the weight of the package by weighing and other information by asking. In addition, customers will be asked to decide whether to send express mail or slow mail. The system will automatically calculate the freight and kitty will ask the customer to pay. After the customer pays, the waybill will be printed for posting on the package.

评阅教师	得分

6. (10 Points) A component-level design is given in the following class diagram. Please assess whether this design is following the basic design principles or not. If not, please give your comments as well as improvement solution.



Tips: This component is composed of Client, Bird, Penguin and Swan. When executing the method letbirdfly (Bird) of Client, it is expected to output the statement "fly free in the air". However, when replacing the base class Bird with subclass Penguin, the output statement "I cannot fly" is inconsistent with the expectation.

评阅教师	得分

7. (20 Points) There is a program to calculate the price of freight. Given pricing rules of freight (货运定价规则) as following:

The package charges are as follows:If the receiving place is within the province and the weight is less than or equal to 25kg, 8 yuan per kilogram for express mail and 4 yuan per kilogram for slow mail.If the receiving place is in another province and the weight is less than or equal to 25kg, the express mail is 12 yuan per kilogram and the slow mail is 8 yuan per kilogram.If the weight is more than 25kg, the overweight part will be charged 2 yuan per kilogram

(1) Draw a Flow Graph with simple condition for this program.

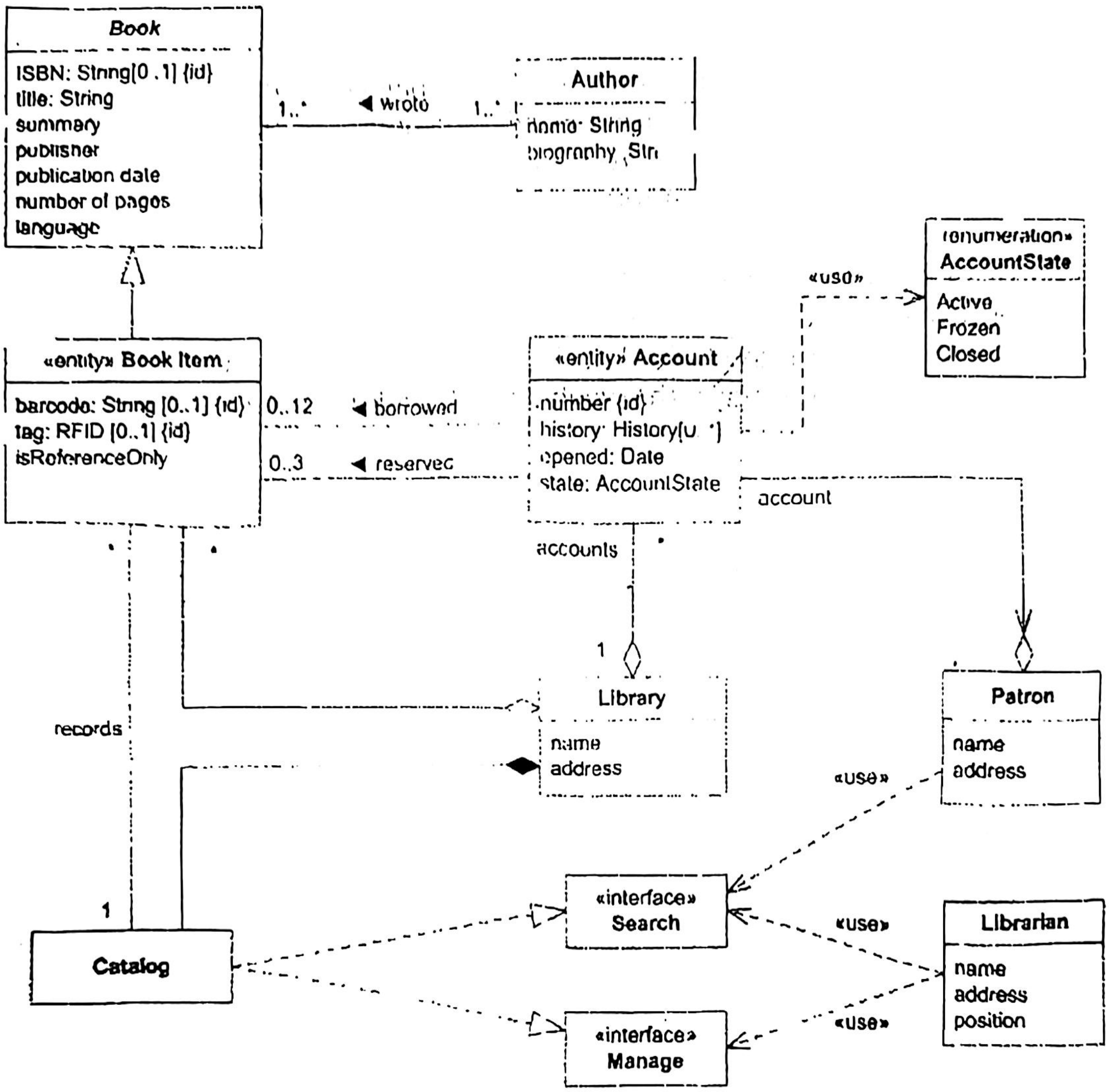
(2) Compute Cyclomatic complexity of the program.

(3) List a set of independent paths for conducting basic path testing.

(4) Design test cases for conducting basic path testing.

评阅教师	得分

8. (10 Points) Following is a Class Diagram of the Digital Library System. Please answer following questions.



(1) What are relationships among class "Catalog", class "Library" and class "Search"?

(2) What is relationship between class "Account" and class "Patron"?

(3) What are relationships among class "Book", class "Author" and class "Book Item"?

(4) What is the multiplicity semantic between class " Book" and class "Author"?

(5) How many types of class are there in this class diagram?

第1题参考答案

原型开发。很多时候，客户提出了软件的一些基本功能，但是没有详细定义功能和特性需求。另一种情况下，开发人员可能对算法的效率、操作系统的兼容性和人机交互的形式等情况并不确定。在这些情况和类似情况下，采用原型开发范型（prototyping paradigm）是最好的解决办法。

虽然原型可以作为一个独立的过程模型，但是更多的时候是作为一种技术，在本章讨论的其他过程模型中应用。不论人们以什么方式运用它，当需求很模糊的时候，原型开发模型可帮助软件开发人员和利益相关者更好地理解究竟需要做什么。

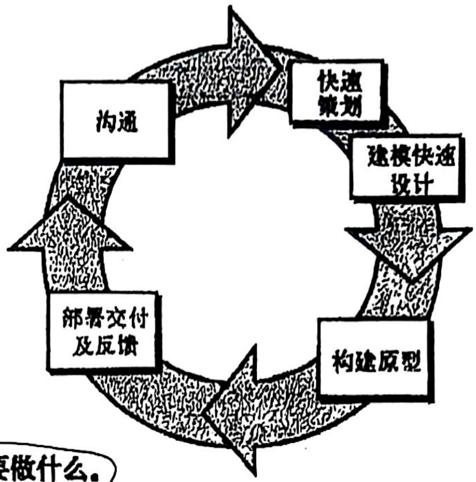


图2-6 原型开发模型

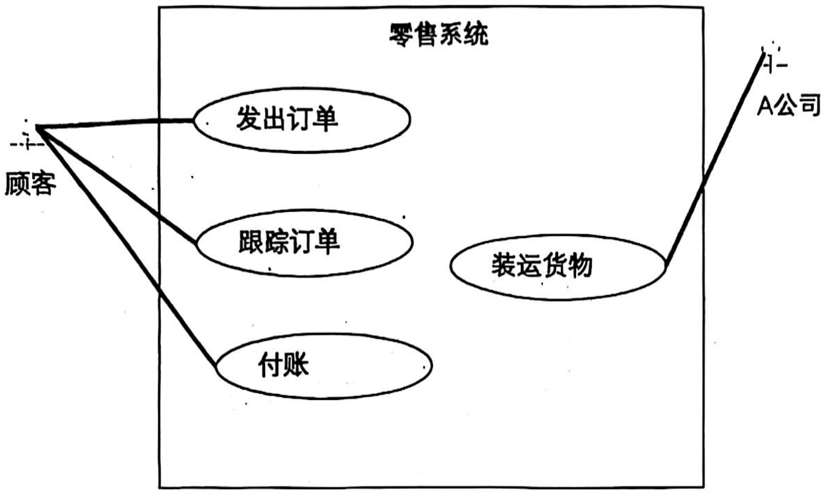
第2题参考答案 单一功能 (2) 高内低耦：功能内聚

相关性

Independence is assessed using two qualitative criteria: cohesion and coupling.
Cohesion is an indication of the relative functional strength of a module.
Coupling is an indication of the relative interdependence among modules.
Cohesion is a natural extension of the information-hiding concept. A cohesive module performs a single task, requiring little interaction with other components in other parts of a program. Stated simply, a cohesive module should (ideally) do just one thing. Although you should always strive for high cohesion (i.e., single-mindedness), it is often necessary and advisable to have a software component perform multiple functions. However, "schizophrenic" components (modules that perform many unrelated functions) are to be avoided if a good design is to be achieved.
Coupling is an indication of interconnection among modules in a software structure. Coupling depends on the interface complexity between modules, the point at which entry or reference is made to a module, and what data pass across the interface. In software design, you should strive for the lowest possible coupling. Simple connectivity among modules results in software that is easier to understand and less prone to a "ripple effect" caused when errors occur at one location and propagate throughout a system.

第3题参考答案

6-

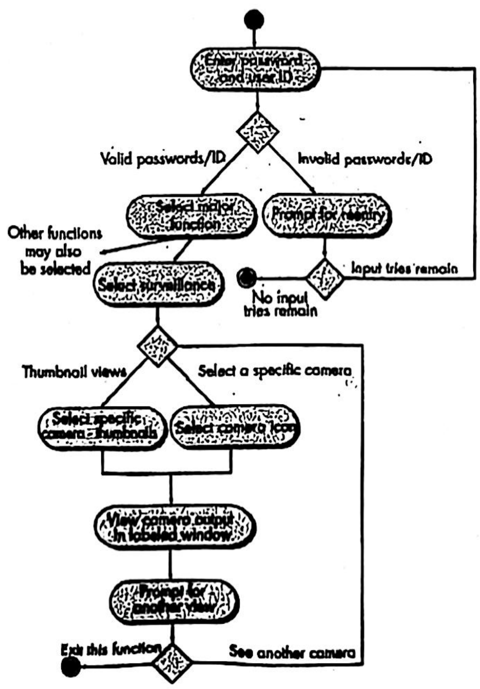


第4题参考答案

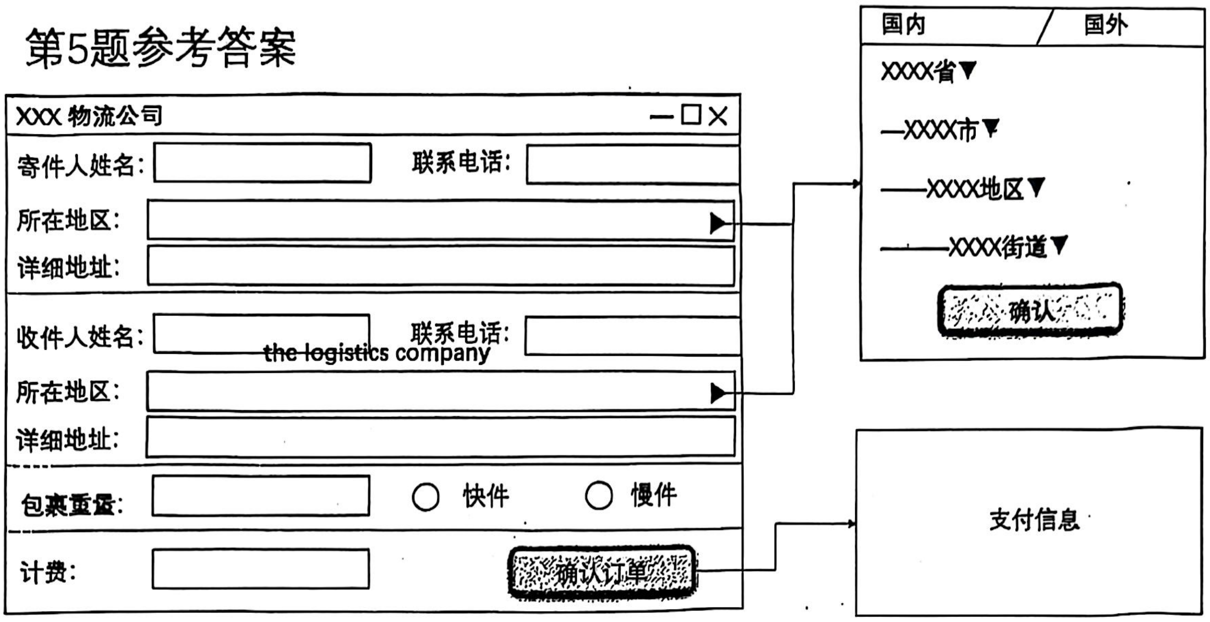
起始一

结束

画成其他图：扣一半

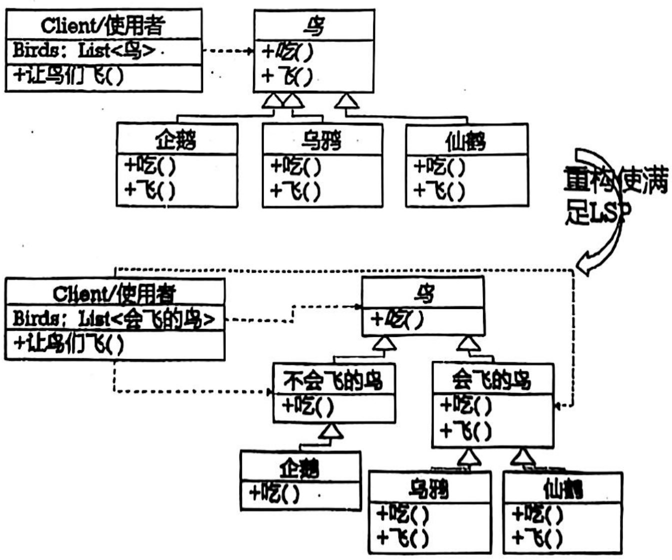


第5题参考答案

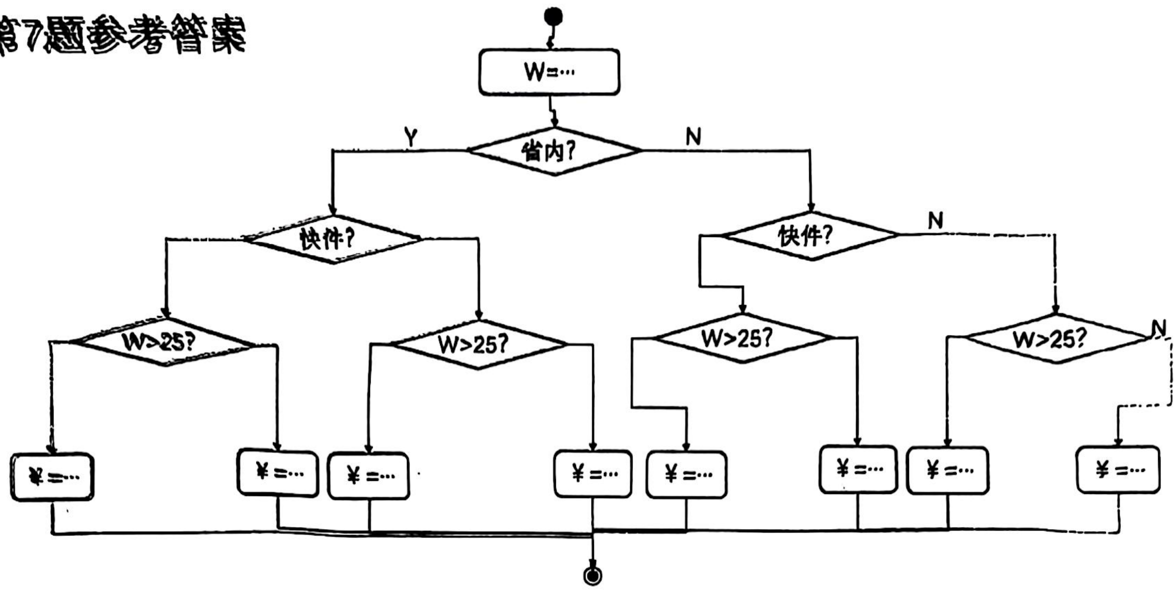


第6题参考答案

违反了LSP：子类可以替换它的基类。
重构如下：



第7题参考答案



第8题参考答案

1. class "Catalog" 和 class "Library" 组合关系 /(聚合关系) 集
class "Catalog"和 interface "Search"实现关系
2. class" Account" and class "Patron"聚合关系 ②
3. class "Book" 和class "Author" 关联关系（Author 写 Book)
class "Book" and class "Book Item"泛化关系（继承关系)
4. What is the multiplicity semantic between class " Book" and class "Author"?
1个作者可以写1本书或多本书，多个作者可以合作写1本书 多对多
5. Book是抽象类，BookItem是实体类，AccountState是枚举类型，Search、Manager是接口，Auhor、Library、Patron、Librarian、Catalog是（实体）类

不用管